

# Specification for NLSv2

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## 1 Introduction

NLSv2 is a synchronous stream cipher with message authentication functionality, submitted to the ECRYPT Network of Excellence call for stream cipher primitives, profile 1A. NLSv2 is an updated version of NLS [19]. The minor change between NLS and NLSv2 increases resistance to attacks utilizing large amounts of keystream. NLS stands for Non-Linear SOBER, and the NLS ciphers are members of the SOBER family of stream ciphers [12],[16],[23] and [24].

NLSv2 is a synchronous stream cipher designed for a secret key that may be up to 128 bits in length. The cipher outputs the key stream in 32-bit blocks. NLSv2 is a software-oriented cipher based on simple 32-bit operations (such as 32-bit XOR and addition modulo  $2^{32}$ ), and references to small fixed arrays. Consequently, NLSv2 is at home in many computing environments, from smart cards to large computers. Source code for NLSv2 is freely available and use of this source code, or independent implementations, is allowed free for any purpose.

NLSv2 includes a facility for simple re-synchronization without the sender and receiver establishing new secret keys through the use of a nonce (a number used only once). This facility does not always need to be used. For example, NLSv2 may be used to generate a single encryption keystream of arbitrary length. In this mode it would be possible to use NLSv2 as a replacement for the commonly deployed RC4 cipher in, for example, SSL/TLS. In this mode, no nonce is necessary. In practice though, much communication is done in messages where multiple encryption keystreams are required. NLSv2 achieves this using a single secret key for the entire (multi-message) communication, with a nonce distinguishing individual messages. NLSv2 is intended to provide security under the condition that no nonce is ever reused with a single key, that no more than  $2^{80}$  words of data are processed with one key, and that no more than  $2^{48}$  words of data are processed with one key/nonce pair. There is no requirement that nonces be random; this allows use of a counter, and makes guaranteeing uniqueness much easier.

This document is arranged as follows. Section 2 introduces the history of NLSv2 and some of the design principles used in the construction. Section 5 contains a complete description of NLSv2. An analysis of the security characteristics, and corresponding design rationale of NLSv2 is found in Section 6.

## 2 Design Considerations

### 2.1 Heritage

Much of the design of NLSv2 can be traced back through the line of SOBER stream ciphers to the original SOBER [23] stream cipher, proposed by Rose in 1998. The algorithm for SOBER is based on 8-bit operations, versus the 32-bit operations used in NLSv2. SOBER was superseded by SOBER-II [24] when various weaknesses were found in the original design. S16 was proposed as 16-bit extension of SOBER-II: S16 copies the structure of SOBER-II and uses 16-bit operations.

However, there were opportunities for strengthening SOBER-II and S16 that could not be ignored. Consequently, replacements for SOBER-II, S16 and a 32-bit version were created. These replacements were called the t-class of SOBER ciphers [12]. The t-class contains three ciphers based on 8-bit, 16-bit and 32-bit operations. The ciphers SOBER-t16 and SOBER-t32 were submitted to the NESSIE program [21]; SOBER-t16 as a stream cipher with 128-bit key strength and SOBER-t32 as a stream cipher with 256-bit key strength. SOBER-t16 and SOBER-t32 proved to be among the strongest stream cipher submissions to NESSIE. However, both ciphers were found to fall short of the stringent NESSIE requirements.

Turing [16] is an adventurous stream cipher proposal that evolved from SOBER-t32. In comparison to other SOBER ciphers, Turing produces five words of output for each internal state update, which is five times more than other SOBER ciphers. The key setup of Turing was found to be lacking [20], but otherwise the design appears strong. However, SOBER proposals since the Turing cipher have returned to the original approach of producing one word of output for each internal state update.

SOBER-128 [17] uses the internal state update function from Turing, but with a very conservative output filter function. SOBER-128 also included an innovative (and flawed [25]) message integrity functionality. Mundja [18], a message integrity primitive based on SHA-256 designed to cooperate with a stream cipher, was developed to rectify the flawed message integrity of SOBER-128.

All these ciphers (SOBER through to SOBER-128) utilize a Linear Feedback Shift Register (LFSR) to produce a changing internal state, with a Non-Linear Filter (NLF) applied to the internal state to produce a word of NLF output. The LFSR provided an internal state with guaranteed cycle period but, on the other hand, the use of an LFSR also enhanced a range of attacks against the ciphers. This suggested that there may be an advantage to replacing the LFSR with a nonlinear feedback shift register. This resulted in NLS [19]: an improved version of SOBER-128 that is based on a nonlinear feedback shift register instead of an LFSR, and with a simplified and more efficient filter function. NLS also incorporates the Mundja primitive for message integrity. NLS was then submitted to the eSTREAM project. NLS was subsequently found to be susceptible to distinguishing attack [5].